

Independent Computational Verification

Paper II — Section 4: clone-class operations (Lemmas 4.1–4.3)

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Artifact: B_clone_class_convexity/audit_B_clone_convexity.py **Verdict:** PASS

Claims verified

- **(B1) Lemma 4.1 (discrete convexity):** for the interpolation family H_k (with k class-vertices attached to N_A and the rest to N_B), the second differences satisfy $f(k-1) + f(k+1) \geq 2f(k)$ where $f(k) = \Phi_\tau(H_k)$; hence $f(k) \leq \max(f(0), f(s))$.
- **(B2) Lemma 4.2:** copying a whole clone class onto another strictly decreases the number of maximal clone classes and **splits no other class**.
- **(B3) Lemma 4.3 + necessity:** copying onto a **simplicial** target preserves chordality; and an explicit example shows a **non-simplicial** target can break it (C_4).

Method

Python 3, NetworkX (graph atlas + chordality), SciPy `linprog` method=`highs`; exact fractions where noted.

Results

- **B1:** configurations tested **8670** (base graphs $|W| \leq 4$, class size ≤ 4 , all neighborhood pairs); minimum second difference **-3.55e-15** (numerically zero), minimum endpoint margin **0.00e+00**. PASS.
- **B2:** admissible class-pairs tested **5663**; clone-class count strictly decreased with no split in every case. PASS.
- **B3:** simplicial-target class-copies tested **5663**; chordality preserved in every case. Necessity example **found**. PASS.

Necessity example (simpliciality is required)

Take the chordal graph G on vertices $\{0, 1, 2, 3\}$ with edges $\{1, 3\}$ and $\{2, 3\}$ (vertex 0 isolated). Copying the source class $\{0\}$ onto the target class $\{3\}$, whose open neighborhood $\{1, 2\}$ is **not** a clique (so the target is non-simplicial), produces the 4-cycle C_4 with edges $\{0, 1\}$, $\{0, 2\}$, $\{1, 3\}$, $\{2, 3\}$, which is **not chordal**. This confirms that the simpliciality hypothesis of Lemma 4.3 is necessary: without it, a clone-class copy can destroy chordality.

Verdict

PASS. Runtime 58.82 s. Discrete convexity, strict progress without splitting, and chordality preservation (with its necessity) all hold across the enumerated universe.

*This document reports an **independent computational verification** of the stated claims: each result was recomputed from scratch by a self-contained script (exact linear programming via SciPy/HiGHS for the fractional triangle-cover number, exact rational arithmetic for the integer optimization, and exhaustive enumeration of non-isomorphic graphs via the NetworkX atlas where*

indicated). This is computational evidence corroborating the analytic proof; it is not itself a formal proof. The machine-checked formal proof is the Lean development, which is conditional on the classical chordal-structure theory.